Maxwell Lokshin

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Education

University of Central Florida Orlando, FL
Bachelor of Science in Computer Science *Jul. 2023 – May 2027*

- Relevant Coursework: Algorithms, Machine Learning, Artificial Intelligence, Computer Vision, Systems Software, Linear Algebra, Discrete Mathematics
- Activities: IEEE @ UCF – SECBOT Software Member; AIRACERS @ UCF – Project Lead; STEEL @ UCF

Experience

Software Engineer Jul. 2025 – Present
PeakActivity *Boynton Beach, FL (Remote)*

- Consult directly with clients to understand technical requirements and contribute to backend solution design
- Build and maintain backend systems applying industry best practices under the guidance of senior engineers
- Learn and apply new coding practices across projects to strengthen full-stack development skills

Projects

Animatronic Scorpion – STEEL @ UCF | *Embedded, ESP32, servo control, Animation* Oct. 2026 – Present

- Writing embedded firmware and a state machine to coordinate 21 servos in synchronized motion for an animatronic scorpion through ESP32

Tubi the Frog – Hackathon | *ESP32, Raspberry Pi, OpenAI, ElevenLabs, Animation* Oct. 2026

- Wrote ESP32 firmware and a custom state machine that animates servo-driven arms and mouth in sync with LLM-generated speech for a conversational animatronic
- Split work across web, Raspberry Pi, and microcontroller over WebSockets and serial; eliminated speaker static and servo noise by separating UART lines
- Used Whipser for speech recognition, Open AI API key for thought process, Eleven Labs for text to speech

Lightning McQueen – F1TENTH | *ROS2, ESP32, CV, LIDAR, Animation* April 2026 – Present

- Built OLED animated eyes from scratch that blink and follow nearby people using an onboard human-detection model
- Developing expressions tied to driving state, onboard audio, and a dead man's switch for an autonomous race car aimed at F1TENTH nationals
- Custom wall following algorithm and speed up/slow down commands that dynamically updates per race track size

TinyTracer – Hackathon (Winner) | *Ray tracing, web simulation* 2026

- Placed 2nd in a OneEthos at a 12-hour hackathon with a team of 3
- Built a custom web-based simulator from scratch where users position solar panels and a ray tracer scores their sunlight capture

Nice Green Machine – F1TENTH | *Python, ROS2, RViz, Docker, Git* Mar. 2026 – Present

- Built hybrid pure pursuit / dual wall-follow algorithm with dynamic mode switching; achieved 9 laps in 5 min at UCF F1TENTH competition
- Set up Gym-ROS simulation in Docker with RViz for pre-hardware validation and integrated VESC messages on Nvidia Jetson TX2

SECBOT – IEEE @ UCF | *C++, Python, ROS2, OpenCV, Gazebo, Docker, Git* Sep. 2025 – Present

- Designed and validated D* Lite path planning in C++ using a grid-based map for efficient field traversal
- Built real-time color vision model for detecting ducks on field during different "missions" (trials)

Technical Skills

Languages: C++, C, Python, TypeScript/JavaScript, Java, SQL (Postgres), HTML/CSS
Robotics & Embedded: ROS2, NAV2, ESP32, Arduino, Raspberry Pi, Nvidia Jetson TX2, servo control, UART, state machines
Web & APIs: React, Node.js/Express, Flask, FastAPI, MongoDB, WebSockets, OpenAI, ElevenLabs, Whisper
Tools & Libraries: Docker, Git, Gazebo, RViz, Fusion 360, OpenCV, NumPy, pandas, Matplotlib, Pygame